Theoretical Computer Science Exam, July 16, 2019

The exam consists of **4 exercises**. Available time: 1 hr 40 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

F.NAME …………………………….. L.NAME ……………………………………….. MATRICOLA …………………

I have passed the midterm exam and I intend to skip the first part (Ex.1-2) (select one): **YES NO**

Exercise 1 (8 pts)

1. Design a least general grammar that generates the language $L = \{ x \in \{a, b\}^* \mid \#_a(x) \bmod 3 = 2 \}$, where $\#_a(x)$ denotes the number of symbols a occurring in the string x, and *mod* is the usual operator that returns the remainder of integer division (e.g., 7 mod 3 = 1, 14 mod 3 = 2, 6 mod 3 = 0).
2. Write a derivation for the string $abba \in L$ according to the above defined grammar.
3. Design a least powerful automaton that accepts language L defined above.
4. Write a computation of the above automaton that accepts the string $abba \in L$
5. What is the least general type of grammar that generates the language $L_1 = L^*$, where L is the language considered in the above points, and the least powerful type of automaton accepting the same language L_1?
6. Consider language $L_2 = \{ x \in \{a, b\}^* \mid \#_a(x) = 2 \cdot \#_b(x) \}$. Design a least powerful automaton that accepts language L_2, and write the computations of the above automaton that accept the strings baa, aba, $aab \in L$

Exercise 2 (8 pts)

Specify, by means of suitable pre- and post-conditions, written in the usual first-order logic for program specification, a subprogram ***somAll* (*v, n, p, s, a*)** where:

– ***v*** and ***n*** are input parameters representing, respectively, an array of integers and the number (greater than 2) of its elements; it is assumed that the integers stored in ***v*** are not all equal; for instance, [1, 2] and [2, 2, 2] are both invalid values for ***v***

– ***p*** is an integer input parameter, such that $p>1$, assumed to be prime

– ***s*** and ***a*** are two output Boolean parameters**,** whose value, after the execution of the *somAll* subprogram, is determined by the following clauses.

  - ***s*** is true if and only if the input prime number ***p*** is an integer divisor of *some* element of ***v***
  - ***a*** is true if and only if the input prime number ***p*** is an integer divisor of *all* the elements of ***v***

Exercise 3 (8 pts)

Consider any family F of sets such that F has the following properties

a. F is not empty
b. F is *closed under complementation* (for every set $S \in F$ also $\neg S \in F$, $\neg S$ being the complement of set S)
c. $\varnothing \notin F$ (all sets in F are non-empty)

For each of the following statements,

1. All sets of the family F are decidable
2. All sets of the family F are semidecidable and not decidable
3. No set of the family F is semidecidable
4. Some set of the family is semidecidable and not decidable

say if it is:

– necessarily true (it holds for every family F with the above features), or
– necessarily false (it does not hold for any family F with the above features), or
– possibly true/false (it holds for some family but not for some other).

Suitably motivate your answer.

Do the answers you provided above change if, in addition, the elements of the family F are sets of indices of computable functions? Suitably motivate your answer.

Exercise 4 (8 pts)

Design a Turing machine that accepts as input a string of type a^n, with $n > 1$, and writes on the output tape the string:

$$c^n b c^{n-1} b^2 c^{n-2} \dots b^{n-2} c^2 b^{n-1} c\, b^n$$

for example if the input is *aaa* then the output is *cccbccbbcbbb.*

Identify the time complezity function of the designed TM and determine its time complexity class. Solutions where the time complexity of the Turing machine is small are preferred.

Briefly elaborate on the time complexity of a RAM machine that computes (in a reasonably efficient way) the same function and compare it with that of the TM, especially in the case when the logarithmic cost is considered for the RAM.

Solutions

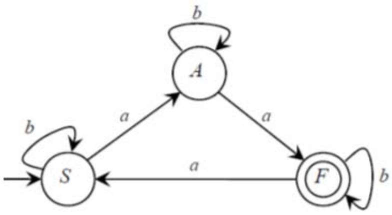
Exercise 1

1. The language is notoriously regular, hence a regular grammar suffices.

   $S \to b\,S \mid a\,A \quad A \to b\,A \mid a\,F \quad F \to b\,F \mid a\,S \mid \varepsilon$

2. $S \Rightarrow a\,A \Rightarrow a\,b\,A \Rightarrow a\,b\,b\,A \Rightarrow a\,b\,b\,a\,F \Rightarrow a\,bb\,a$

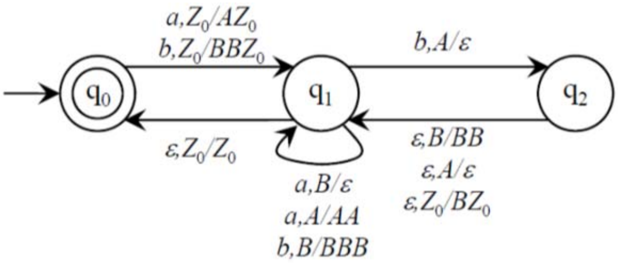
3. In the finite automaton below we use as names of the states those of the nonterminal symbols of the grammar.



4. $S \xrightarrow{a} A \xrightarrow{b} A \xrightarrow{b} A \xrightarrow{a} F$ accept

5. Since language L is regular, also L^* is regular, so that it is generated by a regular grammar and recognized by a finite state automaton.

6. A deterministic PDA suffices



   Here below the computations for the strings baa, aba, $aab \in L$

   $<q_0, Z_0, baa> \vdash <q_1, Z_0BB, aa> \vdash <q_1, Z_0B, a> \vdash <q_1, Z_0, \varepsilon> \vdash <q_0, Z_0, \varepsilon>$ *accept*

   $<q_0, Z_0, aba> \vdash <q_1, Z_0A, ba> \vdash <q_2, Z_0, a> \vdash <q_1, Z_0B, a> \vdash <q_1, Z_0, \varepsilon> \vdash <q_0, Z_0, \varepsilon>$ *accept*

   $<q_0, Z_0, aab> \vdash <q_1, Z_0A, ab> \vdash <q_1, Z_0AA, b> \vdash <q_2, Z_0A, \varepsilon> \vdash <q_1, Z_0, \varepsilon> \vdash <q_0, Z_0, \varepsilon>$ *accept*

Exercise 2

Pre: $n > 2 \wedge \neg\exists\, x\, \forall y\, (\, 1 \le y \le n \;\to\; v[y] = x) \wedge \neg\exists\, q_1\, q_2\, (1 < q_1 < p \;\wedge\; 1 < q_2 < p \;\wedge\; p = q_1 \cdot q_2)$

Post: $(\, s \leftrightarrow \exists\, y\, (\, 1 \le y \le n \;\wedge\; \exists\, z\, (z \ne 1 \wedge z \ne v[y] \wedge\; v[y] = p \cdot z\,)\,)\,) \;\wedge$

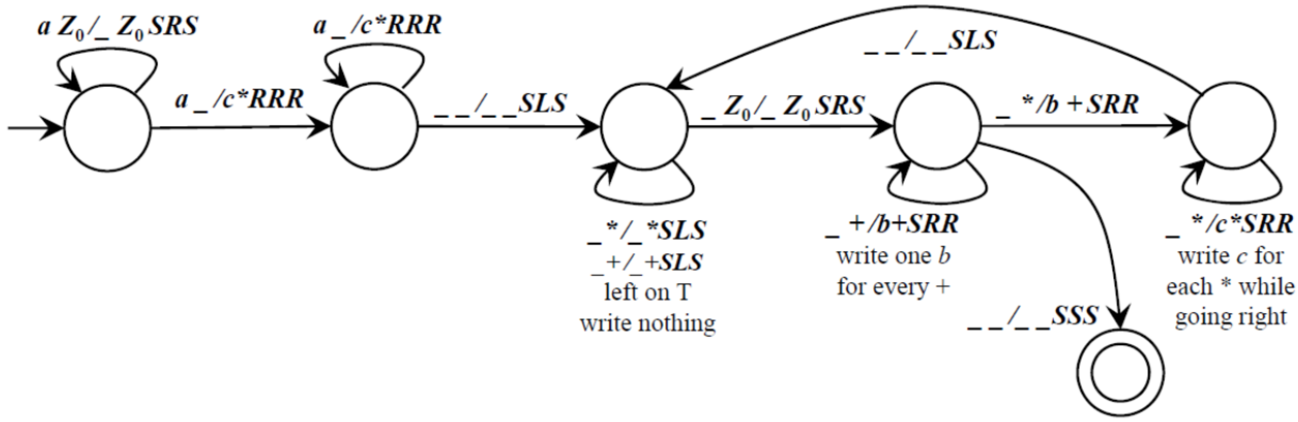
$(\, a \leftrightarrow \forall\, y\, (\, 1 \le y \le n \;\to\; \exists\, z\, (z \ne 1 \wedge z \ne v[y] \wedge\; v[y] = p \cdot z\,)\,)\,)$

Exercise 3

1. All sets of the family *F* are decidable: this can be true for some family and false for some other, because the complement of a decidable set is decidable, hence the complement of an undecidable set is undecidable. A family may include some decidable sets (and their complement sets, that are decidable) and some undecidable sets (and their complement sets, that are undecidable).
2. All sets of the family *F* are semidecidable and not decidable: this is false in all cases, because for every set in the family, its complement set is also in the family, and if a set and its complement are semidecidable, then they are both decidable.
3. No set of the family *F* is semidecidable: possible (for instance, if all the sets of *F* are neither decidable nor semidecidable), but not necessary (the family may include semidecidable sets without violating the properties of *F*).
4. Some set of the family is semidecidable and not decidable: possible (in this case the complement of the semidecidable and not decidable set would not be semidecidable), but not necessary (for instance all sets in *F* may be decidable).

If, in addition, the elements of the family F are sets of indices of computable functions, then case 1 above is impossible (because all the sets of F satisfy the hypothesis of the Rice theorem) and the other cases are unchanged.

Exercise 4



The time complexity function is $T(n) = 1 + n + 1 + (2 \cdot n + 2) \cdot n$, which is $\Theta(n^2)$.

A RAM machine would use some integer counter variables to determine the input length and to write the output. The time complexity class, with the uniform criterion, would be the same, as it derives from the task of writing the output, whose length is n^2. Adopting the logarithmic cost criterion, the managing of the counters would, as usual, introduce an additional logarithmic cost factor that, for large inputs, would make the RAM machine (slightly) less efficient than the TM.